Venkata Lakshmi Parimala Pasupuleti

Staff Software Engineer · Change Management Platform · CI/CD · Observability · Reliability Engineering · Scalable Distributed Systems · Agentic AI · Python · AWS

+1-703-261-5410 | lakshmiparimalap@gmail.com | linkedin.com/in/venkata-lakshmi-parimala-pasupuleti | github.com/venkatalakshmiparimala | Virginia, USA

PROFESSIONAL SUMMARY

Software Engineer with 8 years owning scalable distributed backend systems and platform reliability in production - and a track record of building change management tooling that makes production deployments safe, fast, and observable. Built AgentProbe: a CI-integrated change management platform that blocks quality regressions before merge, enforces API contract stability across producers and consumers, and provides full observability into system behavior across 1,240+ integration test cases. At Entrata, drove end-to-end delivery of 20+ production platform modules at 99.5% uptime across 100K+ monthly active users, led reliability engineering that held bugs below 10 for 6+ consecutive months, and built internal observability tooling reducing triage time 40%. Built agentic workflow integration into backend pipelines (LangGraph, FastAPI). Mentoring 5+ engineers; 300+ code reviews. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, Scalable Distributed Systems, AWS (S3, EC2, RDS, Glue), Azure (OpenAI), Redis, RabbitMQ, REST APIs, API Contract Management, CI/CD (GitHub Actions), Integration Testing, Reliability Engineering, Observability, Incident Response, PostgreSQL, MySQL, Oracle, Linux, Git

Platform & Change Management: Change Management Tooling, Production Deployment Pipelines, Progressive Delivery, Quality Gates, Regression Prevention, Contract Stability, Developer Productivity, Performance Monitoring, Telemetry, On-Call & Operational Excellence

AI & Agents: Agentic Workflows, LangGraph, LangChain, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, HITL, Prompt Engineering, OpenAI API, Anthropic API, MCP, FastAPI

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built scalable agentic workflows (LangGraph stateful graph) for mission planning with CI/CD-integrated validation guardrails blocking non-compliant outputs before delivery - a change management pattern enforcing reliability at every deployment.
- Implemented RAG pipeline with observability on retrieval quality and response groundedness; integration tests across 100% of mission plan compliance scenarios.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built data-intensive ML pipelines (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from ingestion through feature engineering, model optimization, and statistical validation across 190+ countries.
- Mentoring 600+ students in Python, SQL, and database design; designed observability dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Built and owned platform reliability and change management tooling** - internal observability infrastructure reducing incident triage time 40%; reliability engineering maintaining sub-10 bug count for 6+ consecutive months; 99.5% uptime SLA across 120+ Agile sprints at 100K+ monthly active users.
- **Designed scalable distributed backend architecture and production deployment pipelines** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); RabbitMQ event-driven processing with retry logic and idempotency guarantees; progressive delivery patterns across multi-tenant production environments.
- **Owned API contract management across 20+ production modules** - designed and maintained stable API contracts across producers and consumers; integration testing frameworks preventing breaking changes from reaching production; 200+ SQL query optimizations (30% performance gain).
- **Cloud infrastructure at production scale** - AWS S3, EC2, RDS; 100% PHI compliance (Libsodium-encrypted pipelines, zero security incidents); scalable data processing pipelines (fragmented federal data into 180,000 domain records per release cycle).
- Mentoring 5+ junior engineers across technical direction and delivery; 300+ peer code reviews; driving engineering excellence across the team.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management backend modules (PHP, Python, PostgreSQL); scalable data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: CI/CD Change Management & Observability Platform Change Mgmt · CI/CD

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Production change management platform: CI-integrated regression gate blocking quality regressions before merge, API contract stability enforced across producers and consumers, 5-metric observability engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge quality scoring, and multi-provider routing for performance optimization. Integration test harness across 1,240+ test cases - zero regressions shipped to production.

Leasel: Agentic Workflow Backend Platform Agentic · Backend

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Scalable distributed agentic workflow backend: 8-agent stateful graph (LangGraph) with conditional routing, API-driven multi-source data ingestion, and reliability controls - HITL escalation on low-confidence outputs. 93.6% precision, 87.9% F1 across 1,240 integration test cases; CI reliability gate with telemetry-driven quality scoring.

AI on FHIR: NLP-Driven Medical Records Query System NLP · Backend

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Scalable NLP backend and React frontend translating natural-language queries into FHIR API calls - reduced retrieval time ~60%; secure RBAC with full data isolation across structured medical records.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017